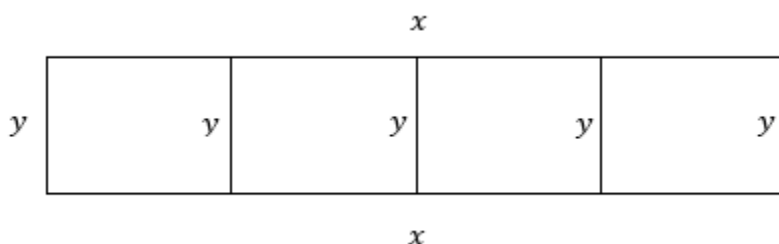
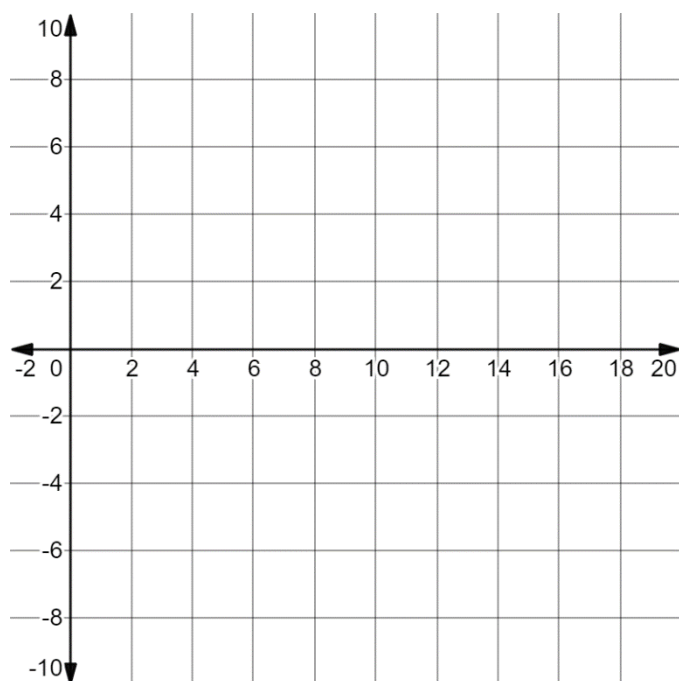


Felix wants to build a rectangular fence in his yard that will be divided into *four* equal pens for chickens. Each pen needs to be the same size. He has 30 feet of fencing to build the fence. This scenario can be modeled as a *two*-variable inequality.

- Describe what the two variables would represent in the context of Felix's fence.
- The diagram below shows a possible model of Felix's fence with sides labeled as x and y .



- Recall Felix had 30 feet of fencing to use. Write a *two*-variable inequality that Felix could use to find possible dimensions for his fence.
- Explain how to determine which inequality symbol to use.
- Graph the inequality.



Drew was applying for a personal loan to pay for home improvement projects. His bank requires a minimum of \$1,000 between his checking and savings accounts to be considered for a loan.

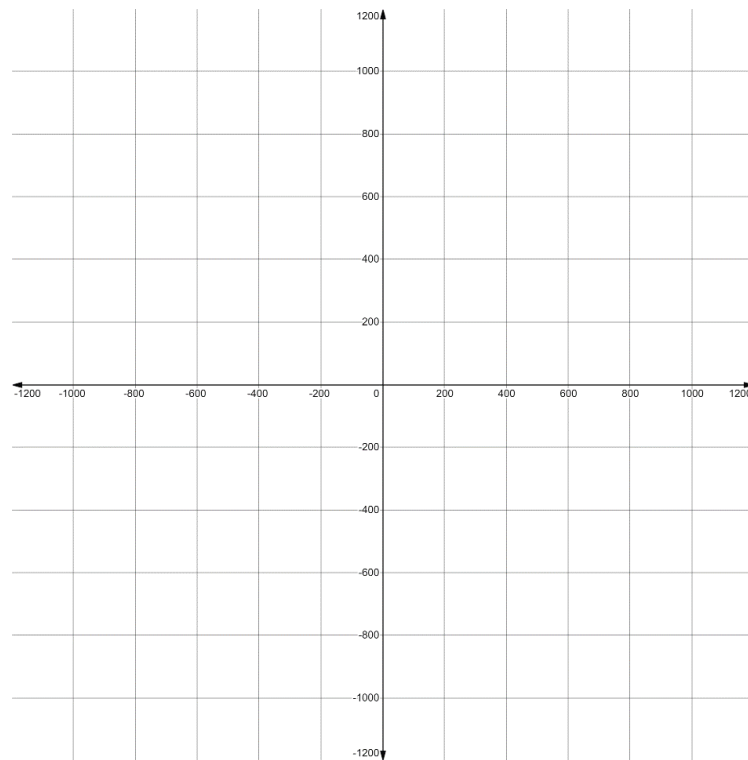
6. Define two variables for this situation.

Let $x =$ _____

Ley $y =$ _____

7. Write an inequality in terms of x and y that could be used to determine whether Drew meets the minimum requirement for the loan.

8. Graph the inequality.



9. Complete the statement.

To be a viable solution in this context, the x and y values must both be rounded to _____ decimal places.

10. Drew overdraw his checking account, which now has a balance of $-\$350.00$. He has $\$1,452.65$ in his savings account. Does he meet the minimum requirement?